

PROJECT REPORT

Data Cleaning and Visualization Using Python on Titanic Dataset

Abstract

This project focuses on cleaning, processing, and visualizing raw data using Python. The Titanic dataset was used for performing data preprocessing tasks such as handling missing values, removing duplicates, and detecting outliers. Python libraries like Pandas, NumPy, Matplotlib, and Seaborn were used for data analysis and visualization. Different graphs and plots were generated to identify patterns and insights from the dataset. This project improved understanding of data preprocessing, exploratory data analysis, and storytelling with data.

Introduction

Data cleaning and visualization are important steps in data analysis and machine learning. Raw datasets often contain missing values, duplicate records, inconsistent data, and outliers which can affect analysis accuracy. Data preprocessing helps improve the quality of the dataset.

Visualization helps in understanding patterns, trends, and relationships within the data. In this project, the Titanic dataset was cleaned and analyzed using Python.

Objectives

The main objectives of this project are:

- To clean raw data using Python
 - To handle missing values and duplicates
 - To detect and remove outliers
 - To visualize data using graphs and plots
 - To generate meaningful insights from the dataset
-

Tools and Technologies Used

Tool / Library	Purpose
Python	Programming Language
Pandas	Data manipulation and cleaning
NumPy	Numerical operations
Matplotlib	Data visualization
Seaborn	Statistical visualization
Google Colab	Code execution platform

Dataset Description

The Titanic dataset contains information about passengers aboard the Titanic ship. The dataset includes passenger details such as age, gender, passenger class, fare, and survival status.

Dataset Features

Column Name	Description
PassengerId	Unique passenger ID
Survived	Survival status
Pclass	Passenger class
Name	Passenger name
Sex	Gender
Age	Passenger age
SibSp	Number of siblings/spouses
Parch	Number of parents/children
Ticket	Ticket number
Fare	Ticket fare
Cabin	Cabin number
Embarked	Port of embarkation

Importing Libraries

The required Python libraries were imported for data cleaning and visualization.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Loading Dataset

The Titanic dataset was loaded using Pandas.

```
url = "https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv"
df = pd.read_csv(url)
```

The first few rows of the dataset were displayed using:

```
df.head()
```

Output of df.head()

FIRST 5 ROWS OF DATASET

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

Dataset Information

Dataset information and statistical summary were checked using:

```
df.info()
df.describe()
```

This step helps identify:

- Missing values
- Data types
- Numerical statistics
- Dataset structure

Output of df.info() and df.describe()

```
DATASET INFORMATION
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId      891 non-null    int64
1   Survived         891 non-null    int64
2   Pclass           891 non-null    int64
3   Name             891 non-null    object
4   Sex              891 non-null    object
5   Age              714 non-null    float64
6   SibSp            891 non-null    int64
7   Parch            891 non-null    int64
8   Ticket           891 non-null    object
9   Fare             891 non-null    float64
10  Cabin            204 non-null    object
11  Embarked         889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

```
STATISTICAL SUMMARY
```

	PassengerId	Survived	Pclass	Age	SibSp	\
count	891.000000	891.000000	891.000000	714.000000	891.000000	
mean	446.000000	0.383838	2.308642	29.699118	0.523008	
std	257.353842	0.486592	0.836071	14.526497	1.102743	
min	1.000000	0.000000	1.000000	0.420000	0.000000	
25%	223.500000	0.000000	2.000000	20.125000	0.000000	
50%	446.000000	0.000000	3.000000	28.000000	0.000000	
75%	668.500000	1.000000	3.000000	38.000000	1.000000	
max	891.000000	1.000000	3.000000	80.000000	8.000000	

	Parch	Fare
count	891.000000	891.000000
mean	0.381594	32.204208
std	0.806057	49.693429
min	0.000000	0.000000
25%	0.000000	7.910400
50%	0.000000	14.454200
75%	0.000000	31.000000
max	6.000000	512.329200

Handling Missing Values

The dataset contained missing values in columns such as Age, Cabin, and Embarked. Certain methods are been used to handle the missing values.

Methods Used:

1. Filling Missing Age Values

```
df['Age'] = df['Age'].fillna(df['Age'].mean())
```

2. Filling Missing Embarked Values

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

3. Removing Cabin Column

```
df.drop('Cabin', axis=1, inplace=True)
```

Output of the Missing Values Before and After Cleaning

MISSING VALUES

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64
```

MISSING VALUES AFTER CLEANING

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age             0
SibSp           0
Parch           0
Ticket          0
Fare            0
Embarked        0
dtype: int64
```

Removing Duplicate Values

Duplicate rows were checked and removed using:

```
print(df.duplicated().sum())  
df.drop_duplicates(inplace=True)
```

This step ensures dataset consistency and improves data quality.

Duplicate Checking Output

```
DUPLICATE ROWS  
0  
  
DUPLICATES REMOVED
```

Outlier Detection and Removal

Outliers in the Fare column were detected using boxplots and removed using the IQR method.

Boxplot Visualization

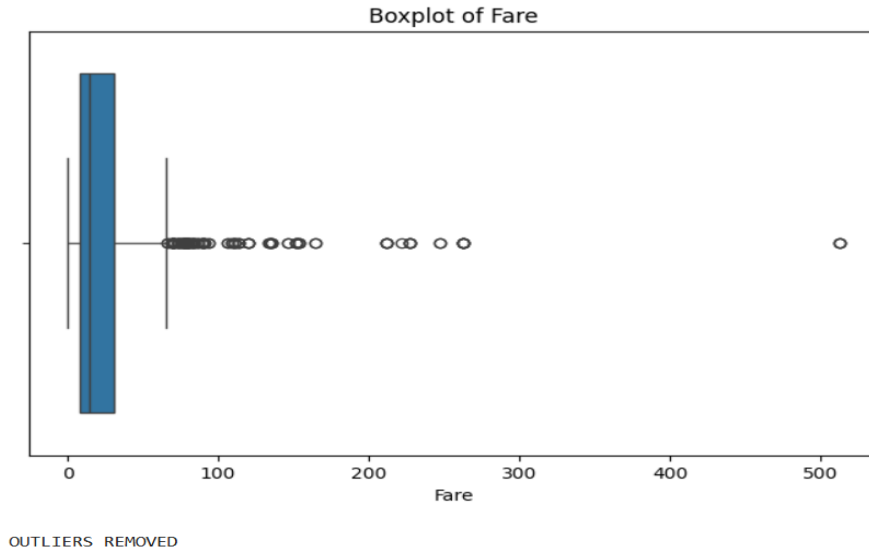
```
sns.boxplot(x=df[ 'Fare' ])  
plt.show()
```

IQR Method

```
Q1 = df[ 'Fare' ].quantile(0.25)  
Q3 = df[ 'Fare' ].quantile(0.75)  
IQR = Q3 - Q1
```

Outliers were removed to improve analysis accuracy.

Boxplot of Fare Column



Data Visualization

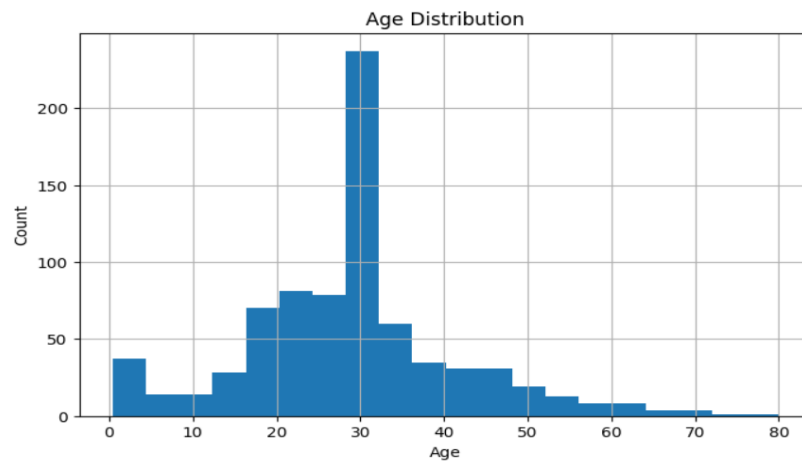
Different visualizations were created to analyze the dataset.

Histogram

A histogram was used to visualize the age distribution of passengers.

```
df['Age'].hist()  
plt.show()
```

Histogram of Age Distribution

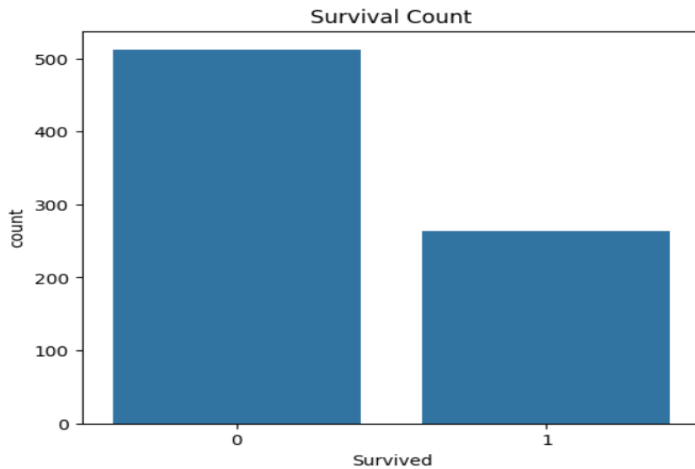


Survival Count Plot

A count plot was used to visualize the number of survivors and non-survivors.

```
sns.countplot(x='Survived', data=df)
plt.show()
```

Survival Count Plot

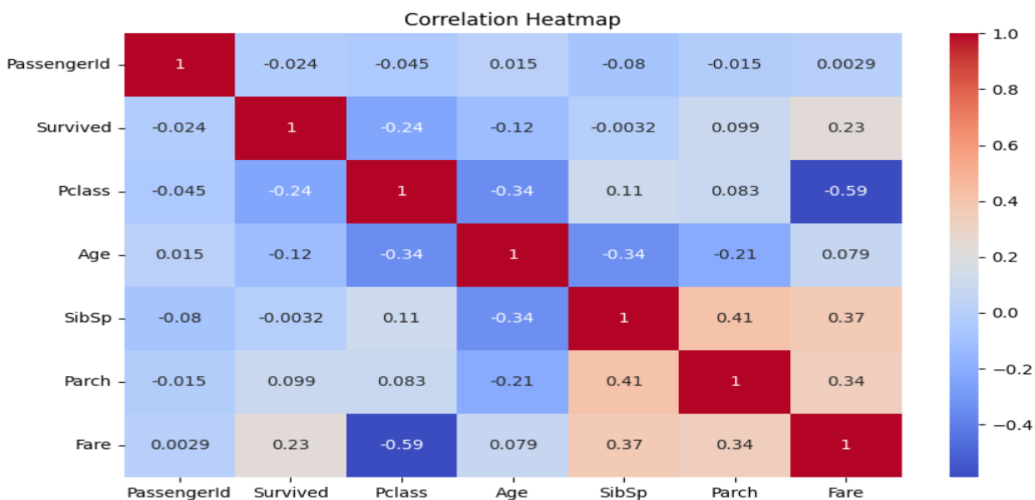


Correlation Heatmap

A heatmap was used to identify relationships between numerical features.

```
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.show()
```

Correlation Heatmap

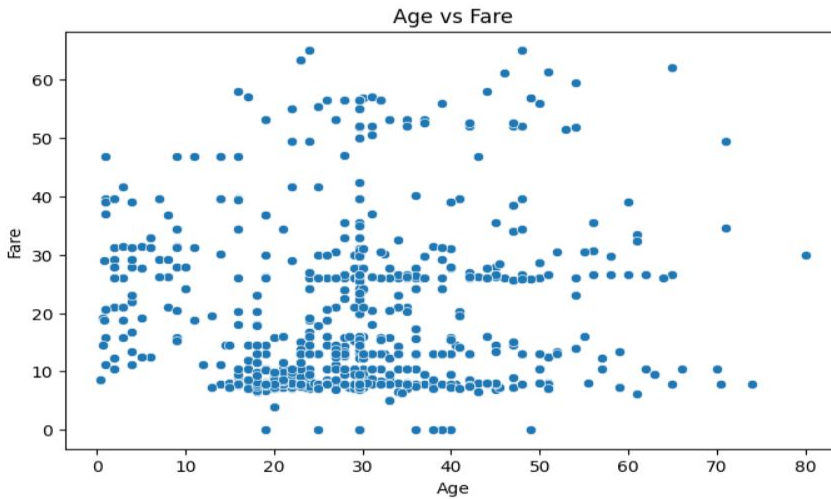


Scatter Plot

A scatter plot was used to analyze the relationship between Age and Fare.

```
sns.scatterplot(x='Age', y='Fare', data=df)
plt.show()
```

Scatter Plot of Age vs Fare

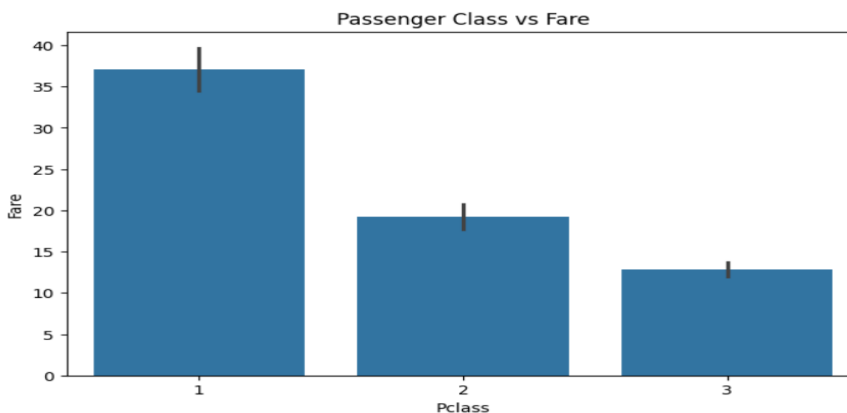


Bar Plot

A bar plot was used to compare passenger class and fare.

```
sns.barplot(x='Pclass', y='Fare', data=df)
plt.show()
```

Passenger Class vs Fare Bar Plot



Saving Cleaned Dataset

The cleaned dataset was saved using:

```
df.to_csv("cleaned_titanic.csv", index=False)
```

This allows future use of the cleaned data.

Final Output of Cleaned Dataset Saved Message

CLEANED DATASET SAVED SUCCESSFULLY

PROJECT INSIGHTS

1. Missing values were successfully handled.
 2. Duplicate rows were removed.
 3. Outliers in Fare column were removed.
 4. Most passengers were aged between 20 and 40.
 5. Higher-class passengers paid higher fares.
 6. Correlation heatmap shows relationships between features.
-

Insights and Findings

The following insights were identified from the analysis:

- Most passengers were between 20 and 40 years old.
 - Higher-class passengers paid higher fares.
 - Some extreme fare values were identified as outliers.
 - The dataset contained missing values which were successfully handled.
 - Visualization helped identify relationships between variables.
-

Conclusion

This project successfully demonstrated data cleaning and visualization techniques using Python. Missing values, duplicates, and outliers were handled effectively to improve data quality. Various visualizations were created to analyze the Titanic dataset and generate insights.

The project improved understanding of:

- Data preprocessing
 - Exploratory Data Analysis (EDA)
 - Visualization techniques
 - Insight generation
 - Storytelling with data
-

Future Enhancements

Future improvements for this project may include:

- Building machine learning models
 - Creating interactive dashboards
 - Performing advanced statistical analysis
 - Deploying the project using Streamlit
-

References

- [Python Documentation](#)
 - [Pandas Documentation](#)
 - [Matplotlib Documentation](#)
 - [Seaborn Documentation](#)
 - [Titanic Dataset](#)
-